SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Case Study

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO3 – Precision	**Assessment:**	Assessment Tool 2 – Case Study
Weightage:	40% of CIA		
CO5 Statement:	Apply N-gram models, smoothing techniques, part-of-speech tagging systems and to evaluate effective syntactic analysis. (BL-3)		
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Section / Batch:	CSE-AI	**Date:**	07-08-2026

Code of Conduct

I N Goutham/192472024 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: N Goutham

Instructions

- Read all three case studies carefully before attempting the questions.
- Provide appropriate justifications and interpretations for every computed result.
- Use NLP concepts, algorithms, and terminology wherever applicable.
- Diagrams, flowcharts, tables, or mathematical formulations may be used to support your answers.
- Duration: 30 minutes.

Section / Topic	Focus	Case Study
N-Gram Language Models and Smoothing Techniques	Bigram and Trigram probability computation, Maximum Likelihood Estimation (MLE), Backoff models, Deleted Interpolation, Entropy calculation, uncertainty analysis, real-time next-word prediction	Case Study 1 – Smart Mobile Keyboard Prediction System
Part-of-Speech (POS) Tagging and Word Classes	Penn Treebank tagset, contextual ambiguity resolution, rule-based tagging, stochastic tagging (HMM), emission and transition probabilities, word class identification, chatbot language understanding	Case Study 2 – AI-Powered Customer Support Chatbot
Transformation-Based Tagging (Brill Tagger)	POS tag correction, transformation rules, contextual tagging, linguistic validation, tag sequence refinement, ambiguity handling	Case Study 3 – News Analytics and POS Tag Correction System
Corpus Statistics and Probabilistic Analysis	Word frequency counting, corpus-based probability estimation, entropy-based uncertainty measurement, tagging confidence evaluation	Case Study 3 – News Analytics and POS Tag Correction System
Integrated Syntactic Analysis and NLP Applications	Application of language models, POS tagging, probabilistic methods, corpus analysis, ambiguity resolution, and decision-making in real-world NLP systems	Case Studies 1, 2, and 3

Annexure A – Case Study

CASE STUDY 1: Smart Mobile Keyboard Prediction System

A smartphone company is developing an AI-powered keyboard application that predicts the next word while users type emails, messages, and social media posts. The language model is trained on the corpus:

"Data science is powerful, data science drives innovation, data science is evolving."

The development team employs Bigram and Trigram Language Models, Backoff Models, Deleted Interpolation, and Entropy Analysis to improve prediction accuracy and handle unseen word sequences.

The following statistics are obtained from the corpus:

- C(data) = 3
- C(data science) = 3
- C(science is) = 2
- C(science drives) = 1
- C(science drives innovation) = 1

The interpolation weights are:

- $\lambda_1 = 0.5$ (Trigram)
- $\lambda_2 = 0.3$ (Bigram)
- $\lambda_3 = 0.2$ (Unigram)

The prediction probabilities after the word "science" are:

- P(is) = 0.66
- P(drives) = 0.33

Questions

1. Compute the Maximum Likelihood Estimate (MLE) of the Bigram Probability P(science | data). Interpret how this probability influences next-word prediction in a mobile keyboard application.
2. A user types the unseen sequence "data science improves". Apply the Backoff Model and estimate the probability using lower-order n-grams. Explain why backoff is essential for real-time predictive text systems.
3. Using Deleted Interpolation, calculate the probability of the sequence "data science is". Discuss how interpolation balances reliability and coverage in sparse datasets.
4. Calculate the Entropy of the distribution P(is)=0.66 and P(drives)=0.33. Interpret the result and evaluate how entropy helps measure uncertainty and prediction confidence in intelligent keyboard systems.

CASE STUDY 1: Smart Mobile Keyboard Prediction System

Q1. Compute the MLE of the Bigram Probability P(science | data)

Scenario Understanding

A smartphone keyboard predicts the next word while users type. The system uses a Bigram Language Model to estimate the probability of a word occurring after another word.

Given Data

- C(data) = 3

- C(data science) = 3

Formula
P(science ∣ data)=C(data)C(data science)
Calculation
P(science ∣ data)=33   P(science ∣ data)=1.0

Analysis:

The probability value 1.0 (100%) indicates that whenever the word "data" appears in the training corpus, it is always followed by "science". Therefore, the keyboard will strongly predict science after the user types data.

Conclusion:

The MLE Bigram Probability is:

P(science ∣ data)=1.0

This high probability improves next-word prediction accuracy in the keyboard application.

Q2. Apply the Backoff Model for "data science improves"

Scenario Understanding

The sequence "data science improves" does not occur in the corpus, making it an unseen trigram. To estimate its probability, the system uses a Backoff Model.

Given Data

Corpus contains:

- data science is
- data science drives innovation

The trigram:

data science improves

is unseen.

Backoff Process

Step 1: Check Trigram

P(improves ∣ data science)

Not found in corpus.

Step 2: Backoff to Bigram

P(improves ∣ science)

Not found.

Step 3: Backoff to Unigram

P(improves)

Since "improves" does not occur in the corpus, the probability is very small (approaching 0).

Analysis:

Without backoff, the probability would become zero, causing the prediction system to fail. Backoff allows the model to use lower-order N-grams whenever higher-order N-grams are unavailable.

Conclusion

The sequence receives a small non-zero estimate through lower-order models. Backoff is essential because it handles unseen word sequences and improves real-time predictive text performance.

Q3. Deleted Interpolation for "data science is"

Scenario Understanding

Deleted Interpolation combines trigram, bigram, and unigram probabilities using weights to improve prediction reliability.

Given Data

- $\lambda_1 = 0.5$ (Trigram)
- $\lambda_2 = 0.3$ (Bigram)
- $\lambda_3 = 0.2$ (Unigram)

Formula

$$P(w3 \mid w1, w2) = \lambda 1 \; Ptri + \lambda 2 \; Pbi + \lambda 3 \; Puni$$

Probability Values

For **data science is**

Trigram probability:

$$P(\text{is} \mid \text{data science}) = 32 = 0.67$$

Bigram probability:

$$P(\text{is} \mid \text{science}) = 32 = 0.67$$

Unigram probability:

$$P(\text{is}) = 0.66$$

Calculation:

$P=0.5(0.67)+0.3(0.67)+0.2(0.66)$ $P=0.335+0.201+0.132$ $P=0.668$

Final Probability:

$P(\text{data science is})=0.668$

Analysis:

Interpolation balances:

- Reliability of trigram information.
- Coverage of bigram information.

- Robustness of unigram information.

Thus, even when some N-gram counts are sparse, the system still produces a reliable prediction.

Conclusion

Deleted Interpolation provides a more stable probability estimate than relying on a single N-gram model.

Q4. Entropy Calculation

Scenario Understanding

Entropy measures uncertainty in next-word prediction. Lower entropy means higher confidence, while higher entropy means greater uncertainty.

Given Data

- P(is) = 0.66
- P(drives) = 0.33

Formula

$H(X) = -\sum P(x)\log_2 P(x)$

Calculation

$H = -(0.66\log_2 0.66 + 0.33\log_2 0.33)$

$H = -(0.66(-0.599) + 0.33(-1.599))$

$H = 0.395 + 0.528$

$H = 0.923$ bits

Final Entropy

$H(X) = 0.923$ bits

Analysis:

- Entropy close to 0 → very confident prediction.
- Entropy close to 1 → moderate uncertainty.
- Here 0.923 bits indicates that "is" is more likely than "drives", but there is still some uncertainty.

Conclusion:

Entropy helps the keyboard system measure prediction confidence. Lower entropy results in more reliable next-word suggestions and better user experience.

CASE STUDY 2: AI-Powered Customer Support Chatbot

A multinational airline deploys a customer support chatbot that processes user messages and automatically identifies the grammatical role of words before generating responses.

Consider the following user inputs:

1. "Book a flight ticket now."
2. "This book is interesting."

The chatbot uses the Penn Treebank POS Tagset and a Hidden Markov Model (HMM) for probabilistic tagging.

Given:

- P(book | VB) = 0.6
- P(book | NN) = 0.4
- P(Start → VB) = 0.5
- P(Start → NN) = 0.5

The system must determine whether the word "book" functions as a Verb (VB) or Noun (NN) depending on context.

Questions

1. Assign appropriate Penn Treebank POS tags to every word in both sentences. Justify the tag assigned to the word "book" using contextual clues.
2. Using the HMM probabilities, compute the likelihood of "book" being tagged as a Verb (VB) in the sentence "Book a flight ticket now." Explain why the probabilistic model favors this tag.
3. Compare Rule-Based Tagging and Stochastic (HMM) Tagging in resolving lexical ambiguity. Recommend which approach is more suitable for large-scale chatbot deployment and justify your answer.
4. Evaluate the role of standardized POS tagsets and word classes in improving intent detection, response generation, and overall chatbot performance.

CASE STUDY 2: AI-Powered Customer Support Chatbot

Q1. Assign Penn Treebank POS Tags to Both Sentences

Scenario Understanding

The chatbot must identify the grammatical role of each word before generating responses. The word "book" is ambiguous because it can function as both a Verb (VB) and a Noun (NN) depending on context.

Sentence 1

"Book a flight ticket now."

Word	POS Tag	Meaning
Book	VB	Verb (command/action)
a	DT	Determiner
flight	NN	Noun
ticket	NN	Noun
now	RB	Adverb

Tagged Sentence

Book/VB a/DT flight/NN ticket/NN now/RB

Justification

The sentence begins with an imperative command asking the user to perform an action. Therefore, book functions as a Verb (VB).

Sentence 2

"This book is interesting."

Word	POS Tag	Meaning
This	DT	Determiner
book	NN	Noun
is	VBZ	Verb
interesting	JJ	Adjective

Tagged Sentence
This/DT book/NN is/VBZ interesting/JJ
Justification

The word book follows the determiner this, indicating that it is the object being discussed. Hence, it functions as a Noun (NN).

Conclusion

Context determines whether book is tagged as a Verb (VB) or Noun (NN), demonstrating lexical ambiguity resolution in POS tagging.

Q2. HMM Probability Calculation for "Book a flight ticket now."
Given Data

- P(book | VB) = 0.6
- P(book | NN) = 0.4
- P(Start → VB) = 0.5
- P(Start → NN) = 0.5

Formula

$P(VB)=P(Start \rightarrow VB) \times P(book \mid VB)$

Calculation

$P(VB)=0.5 \times 0.6$ $P(VB)=0.30$

For Noun:

$P(NN)=0.5 \times 0.4$
$P(NN)=0.20$

Comparison

Tag	Probability
VB	0.30
NN	0.20

Analysis:

Since:

$$\mathbf{0.30 > 0.20}$$

the HMM chooses **VB**.

Conclusion

The probabilistic model favors Verb (VB) because it has a higher combined transition and emission probability. Therefore, book is tagged as a verb in the sentence "Book a flight ticket now."

Q3. Rule-Based Tagging vs Stochastic (HMM) Tagging

Scenario Understanding

The chatbot must resolve lexical ambiguities such as the word **book** occurring as both a noun and a verb. Two approaches are available: Rule-Based Tagging and Stochastic (HMM) Tagging.

Comparison

Feature	Rule-Based Tagging	HMM Tagging
Method	Uses handcrafted grammar rules	Uses probabilities
Context Handling	Limited	Strong
Ambiguity Resolution	Difficult	Effective
Scalability	Low	High
Adaptability	Requires manual updates	Learns from corpus
Accuracy	Moderate	High

Analysis:

Rule-Based Tagging

- Uses predefined linguistic rules.
- Works well for small datasets.
- Difficult to maintain for large chatbot systems.

HMM Tagging

- Uses transition and emission probabilities.
- Handles ambiguous words effectively.
- Scales to large datasets.
- Adapts better to real-world conversations.

Recommendation

For a large-scale airline chatbot, Stochastic (HMM) Tagging is preferred because it handles ambiguity automatically and provides higher accuracy.

Conclusion

HMM tagging is more suitable for chatbot deployment due to its probabilistic reasoning, scalability, and ability to process large conversational datasets efficiently.

Q4. Role of Standardized POS Tagsets and Word Classes

Scenario Understanding

The chatbot relies on POS tagging to understand user intent and generate meaningful responses. Standardized tagsets such as the Penn Treebank improve consistency and interpretation.

Importance of POS Tagsets

1. Intent Detection

POS tags help identify user actions and requests.

Example:

Book/VB a flight

The tag **VB** indicates a booking request.

2. Response Generation

Correct tagging allows the chatbot to generate relevant responses.

Example:

book/NN

Refers to an object.

book/VB

Refers to an action.

3. Ambiguity Resolution

The same word can have different meanings.

Example:

Book/VB a ticket

vs

This book/NN is interesting

POS tagging resolves this ambiguity.

4. Improved Chatbot Performance

Benefits include:

- Better language understanding
- Accurate intent classification
- Reduced misunderstanding
- Faster response generation
- Higher customer satisfaction

Analysis:

Standardized POS tagsets provide a common linguistic framework that enables consistent tagging across different datasets and applications. This improves NLP model reliability and overall chatbot efficiency.

Conclusion:

Penn Treebank POS tags and word classes play a critical role in intent detection, ambiguity resolution, and response generation. Their use significantly enhances the accuracy, consistency, and performance of AI-powered customer support chatbots.

CASE STUDY 3: News Analytics and POS Tag Correction System

A digital news analytics company processes millions of news articles daily. The platform uses a Transformation-Based Tagger (Brill Tagger) to improve tagging accuracy after an initial POS tagging phase.

The sentence processed by the system is:

"Economic growth increases employment."

Initial POS Tags:

- economic/JJ
- growth/NN
- increases/NNS
- employment/NN

A learned transformation rule states:

Change NNS to VBZ if the preceding word is tagged as NN.

The system also performs corpus-based frequency analysis and uncertainty evaluation.

Questions

1. Apply the transformation rule and update the POS tag sequence. Explain why the correction is linguistically appropriate.
2. Analyze the initial tagging errors and justify the corrected tag assignments using grammatical structure and sentence semantics.
3. Assuming the corpus contains the words:
    - economic (120 occurrences)
    - growth (450 occurrences)
    - increases (210 occurrences)
    - employment (380 occurrences)

    Compute the word frequency distribution and explain how frequency information supports probabilistic POS tagging.

4. Evaluate how Transformation-Based Tagging improves tagging accuracy compared with the initial tagging stage. Discuss how entropy can be used to measure uncertainty in tag assignments before and after rule application.

CASE STUDY 3: Social Media Trend Analysis System

Q1. Assign Initial POS Tags and Identify Errors

Scenario Understanding

A social media trend analysis system processes millions of posts daily. The sentence:

"They can trend stories quickly."

contains ambiguity because the word **"can"** may function as a modal verb or a main verb. The system initially tags words using statistical methods and later corrects errors using Transformation-Based Tagging (Brill Tagging).

Initial POS Tags

Word	Initial POS Tag
They	PRP
can	NN
trend	VB
stories	NNS
quickly	RB

Tagged Sentence

They/PRP can/NN trend/VB stories/NNS quickly/RB

Error Identification

The word **"can"** is incorrectly tagged as **NN (Noun)**.

In the given sentence, **can** functions as a **Modal Verb (MD)** because it expresses possibility or ability.

Correct POS Tags

Word	Correct POS Tag
They	PRP
can	MD
trend	VB
stories	NNS
quickly	RB

Correct Tagged Sentence

They/PRP can/MD trend/VB stories/NNS quickly/RB

Conclusion

The initial statistical tagger incorrectly assigns **NN** to **can**. Contextual analysis reveals that **can** functions as a modal verb (**MD**).

Q2. Apply Brill Transformation Rule
Given Rule

NN → MD

if previous tag = PRP

and next word = Verb

Initial Tagged Sentence:

They/PRP can/NN trend/VB stories/NNS quickly/RB

Rule Application:

For the word **can**:

- Previous tag = PRP (**They**)
- Current tag = NN
- Next word = trend (Verb)

All conditions are satisfied.

Transformation

can/NN
↓
can/MD

Updated Tagged Sentence

They/PRP can/MD trend/VB stories/NNS quickly/RB

Analysis:

The Brill rule successfully identifies the contextual environment and changes the incorrect noun tag to a modal verb tag.

Conclusion:

Transformation-Based Tagging improves tagging accuracy by applying context-sensitive correction rules after initial tagging.

Q3. Why Transformation-Based Tagging Improves Accuracy

Scenario Understanding

Statistical taggers sometimes assign incorrect tags because they rely only on probability estimates. Transformation-Based Tagging combines statistical tagging with linguistic rules to improve accuracy.

Advantages of Transformation-Based Tagging

1. Context-Aware Corrections

Brill Tagger uses neighboring words and tags to correct mistakes.

Example:

They/PRP can/NN trend/VB

becomes

They/PRP can/MD trend/VB

2. Reduces Ambiguity

Words like:

- can
- book
- record
- present

can belong to multiple POS categories.

Transformation rules resolve ambiguity using context.

3. Combines Statistical and Linguistic Knowledge

- Initial tags from statistical model.

- Corrections from hand-crafted or learned rules.

This produces more reliable results.

4. Improves NLP Performance

Benefits include:

- Better sentiment analysis
- Better information extraction
- Better trend detection
- Better language understanding

Conclusion:

Transformation-Based Tagging increases POS tagging accuracy by correcting errors made by statistical taggers using contextual linguistic rules. This leads to more accurate analysis of social media content.

Q4. Impact of Accurate POS Tagging on Trend Analysis

Scenario Understanding

The social media trend analysis system identifies emerging topics and user interests from large volumes of posts. Accurate POS tagging is essential for understanding the meaning of words and relationships between them.

Importance of Accurate POS Tagging

1. Better Trend Identification

Correctly tagged nouns and verbs help identify trending topics.

Example:

trend/VB stories/NNS

indicates an action related to stories.

2. Improved Information Extraction

POS tags help extract:

- People
- Events
- Products
- Locations
- Activities

from social media posts.

3. Enhanced Sentiment Analysis

Accurate tagging helps identify:

- Adjectives

- Adverbs
- Verbs

that contribute to sentiment.

4. Reduced Misclassification

Incorrect tags may lead to:

- Wrong trend detection
- Incorrect topic clustering
- Poor recommendation quality

Transformation-based correction minimizes these errors.

Analysis:

Accurate POS tagging provides a strong linguistic foundation for downstream NLP tasks such as trend analysis, topic modeling, sentiment analysis, and recommendation systems. By reducing ambiguity and improving grammatical understanding, the system can generate more reliable insights from social media data.

Conclusion:

Accurate POS tagging significantly improves social media trend analysis by enabling correct interpretation of user-generated content. Transformation-Based Tagging further enhances performance by correcting tagging errors and ensuring high-quality linguistic analysis

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Q. No. / Criteria Level	Understanding of Case	Application of Concepts	Analysis	Solution Design	Clarity	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
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Date: ____________________	Date: ____________________	Date: ____________________